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Mobile Communications

Unit 3 Lab Report

GIT LINK: https://github.com/sticker2024/MC_LAB3

Mini Projects 3.1, 3.2 and 3.3: Frequency Reuse, Capacity Planning, Handover and Connection State Machine

1. Introduction

This report presents three simulation mini projects covering the quantitative ideas of Unit 3. Project 3.1 builds a hexagonal frequency-reuse generator and a handover decision simulator. Project 3.2 builds a cellular capacity planning tool. Project 3.3 models a device's connection and handover lifecycle as a finite-state machine. All results and screenshots in this report come directly from the output of the Python scripts mp3_1.py, mp3_2.py and mp3_3.py, which use fixed random seeds so that every number can be reproduced.

Terminal screenshots show the printed output of each script. Figures are the plots saved by the scripts.

2. Mini Project 3.1: Frequency Reuse and Handover Decision Logic

2.1 Objective

To generate hexagonal frequency-reuse layouts and compute the co-channel reuse ratio $Q = D/R = \sqrt{(3N)}$, and to simulate a handover decision with a naive trigger, a hysteresis-only trigger and a hysteresis plus time-to-trigger (TTT) trigger, showing the ping-pong effect and how tuning suppresses it.

2.2 Method

- Hexagon centres are computed in axial coordinates for a pointy-top grid of radius R. Each cell is labelled with a group from 1 to N using cluster offset vectors $v1 = (i, j)$ and $v2 = (-j, i + j)$, where $N = i^2 + ij + j^2$.
- Layouts are drawn for N = 1, 3, 4 and 7 (and N = 9 and 12 as an extension). The script measures the smallest distance between same-group cells and asserts that it equals $\sqrt{(3N)} \cdot R$.
- Serving and neighbour RSRP series (300 samples) are generated from a linear trend, a random walk ($\sigma = 0.25$ dB per step) and Gaussian noise ($\sigma = 3$ dB). The trends cross at sample 150.
- The same trigger function is run with (margin 0 dB, TTT 1), (3 dB, TTT 1) and (3 dB, TTT 5). A streak counter resets whenever the entering condition breaks.

2.3 Results

```
NumPy      : 2.5.3
Matplotlib : 3.11.2

Q versus N (required set)
  N |  Q = sqrt(3N) |  measured D/R | groups seen | cells drawn
--------------------------------------------------------------
  1 |         1.732 |         1.732 |           1 |          37
  3 |         3.000 |         3.000 |           3 |          91
  4 |         3.464 |         3.464 |           4 |          91
  7 |         4.583 |         4.583 |           7 |         169
Layout check passed: measured co-channel distance equals sqrt(3N)*R for every N.

Q versus N (Lab Task 1 extension)
  N |  Q = sqrt(3N) |  measured D/R | groups seen | cells drawn
--------------------------------------------------------------
  9 |         5.196 |         5.196 |           9 |         169
 12 |         6.000 |         6.000 |          12 |         217

Handover comparison (seed 7, 300 samples)
Configuration                       | Handovers | Min serving RSRP | Samples < -100 dBm
-------------------------------------------------------------------------------------------
Naive (margin 0 dB, no TTT)         |        11 |        -102.7 dBm |                  4
Hysteresis only (3 dB, no TTT)      |         5 |        -103.4 dBm |                  5
Hysteresis 3 dB + TTT 5 samples     |         1 |        -105.9 dBm |                 25

Handover sample indices:
  Naive (margin 0 dB, no TTT): [105, 107, 109, 110, 126, 129, 130, 137, 138, 143, 145]
  Hysteresis only (3 dB, no TTT): [105, 107, 109, 110, 126]
  Hysteresis 3 dB + TTT 5 samples: [154]

Too-late check (hyst+TTT): nominal trend crossover at sample 150; first handover from sample 100 on = 154;
      serving RSRP fell below -100 dBm in 25 samples (min -105.9 dBm).

Sweep (Lab Task 3): naive baseline mean 18.6 handovers, 2.1 low-signal samples.
Low-signal budget = baseline + 8 samples = 10.1.
Best feasible setting: margin 0 dB, TTT 6 samples -> 1.00 handovers, 6.2 low-signal samples.
Mean handover count / low-signal samples for a few settings:
  margin 0 dB, TTT  1: 18.60 handovers,    2.1 low samples
  margin 3 dB, TTT  1:  6.75 handovers,    2.8 low samples
  margin 3 dB, TTT  5:  1.00 handovers,    9.0 low samples
  margin 6 dB, TTT 10:  1.00 handovers,   23.1 low samples
  margin 8 dB, TTT 15:  1.00 handovers,   37.0 low samples

Figures and tables written to /home/claude/mc_unit3/out_3_1
```

Figure 1: Terminal output of mp3_1.py (Q table, handover table, sweep summary)

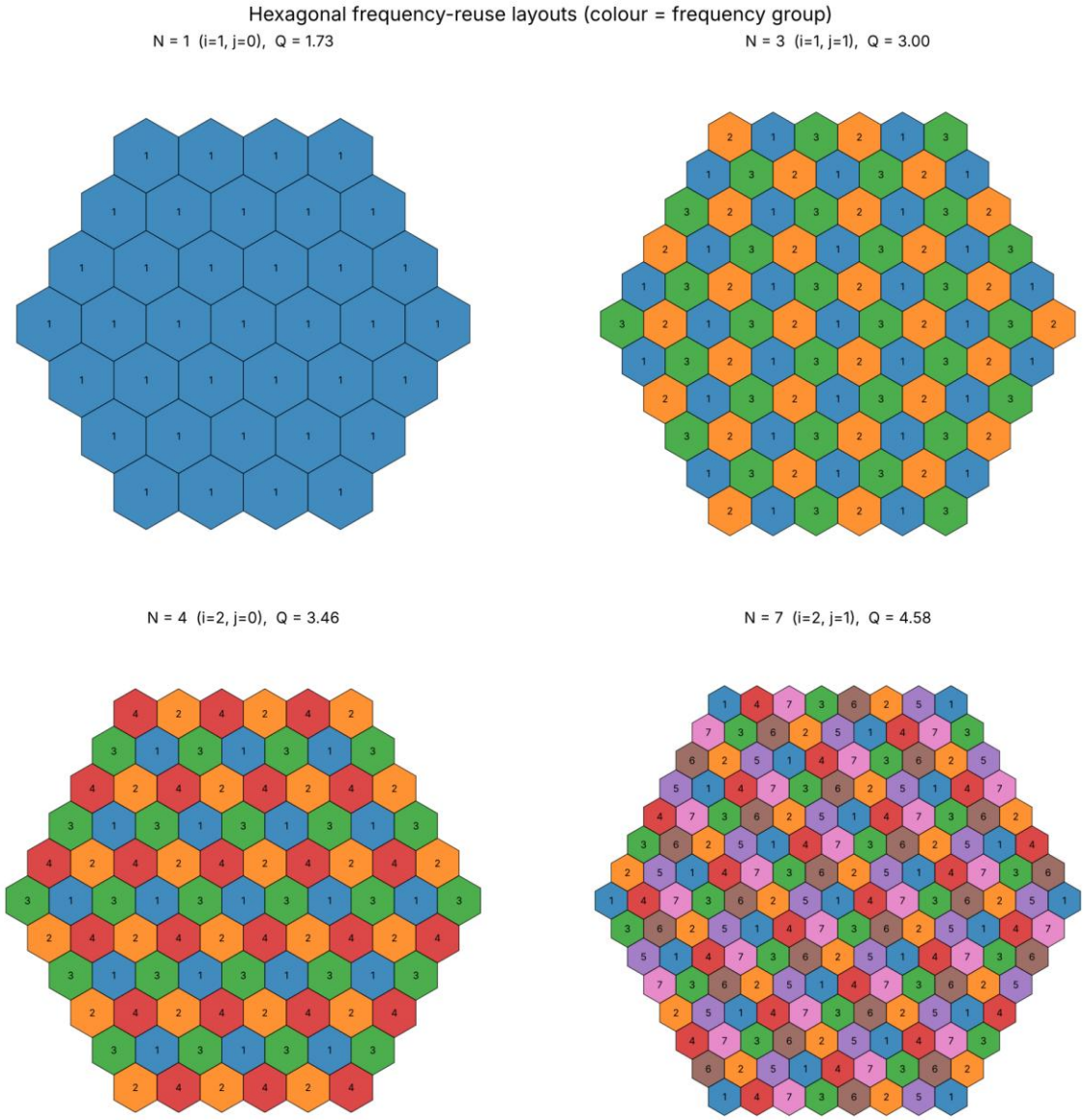


Figure 2: Frequency-reuse layouts for N = 1, 3, 4 and 7 (colour = frequency group)

Table 2: Q versus N (measured nearest same-group distance equals √(3N)·R)

N	i, j	Q = √(3N)	Measured D/R	Groups seen
1	1, 0	1.732	1.732	1
3	1, 1	3.000	3.000	3
4	2, 0	3.464	3.464	4
7	2, 1	4.583	4.583	7
9 (ext.)	3, 0	5.196	5.196	9
12 (ext.)	2, 2	6.000	6.000	12

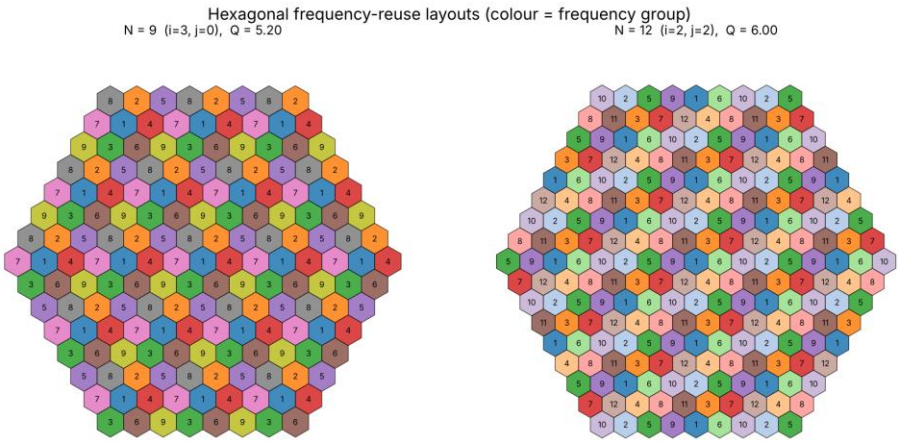


Figure 3: Extension: layouts for N = 9 and N = 12

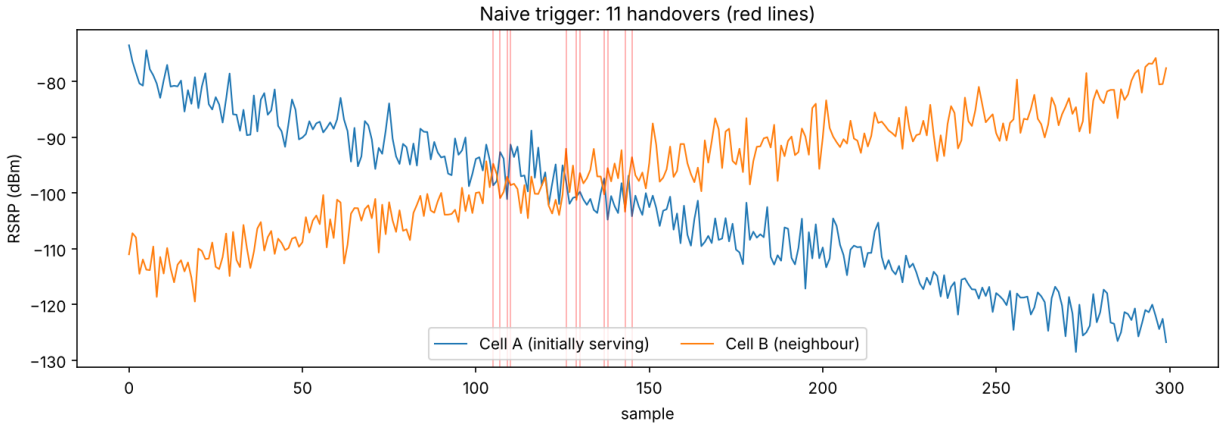


Figure 4: Naive trigger on the synthetic RSRP series (red lines = handovers)

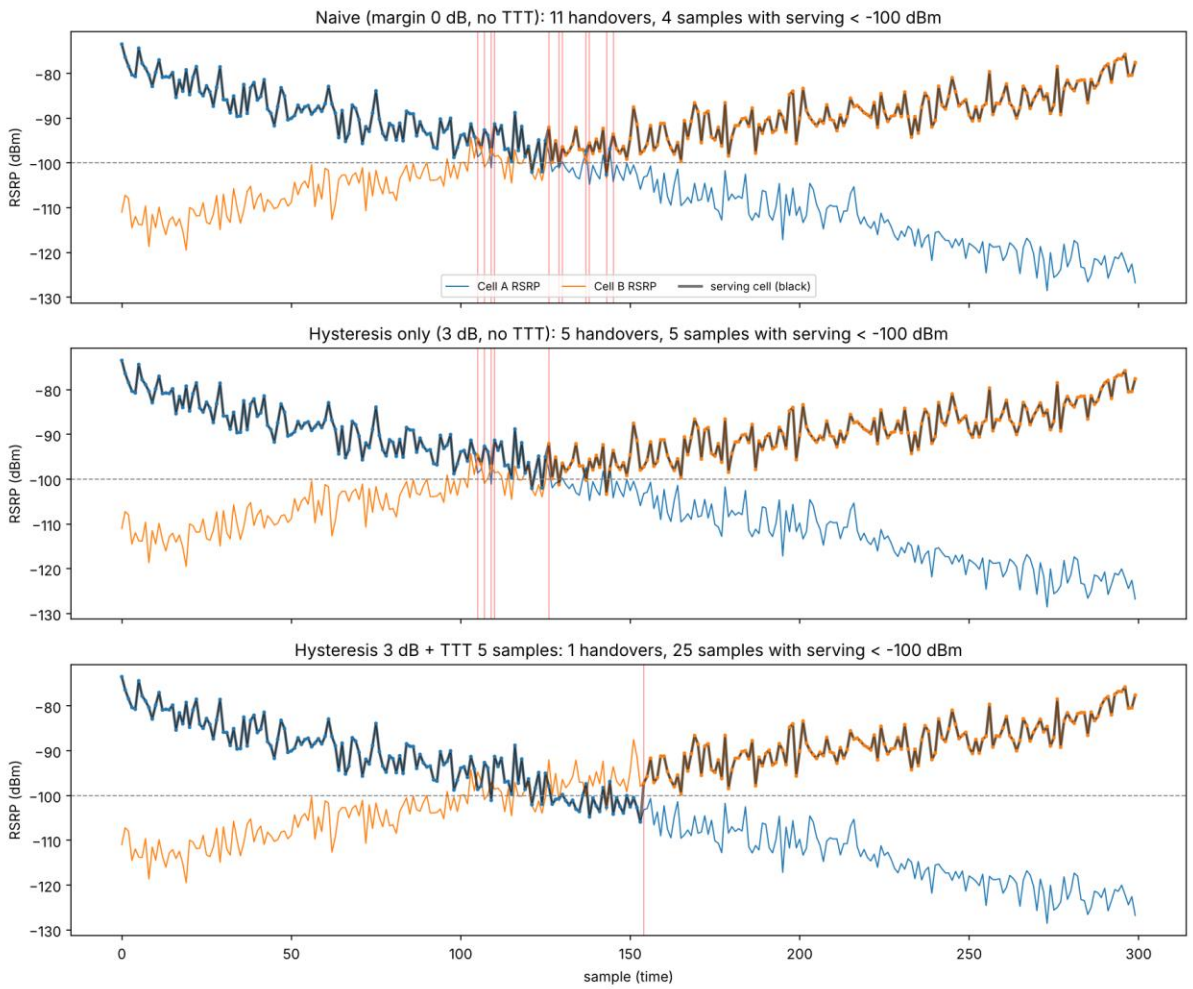


Figure 5: Naive, hysteresis-only and hysteresis plus TTT triggers on the same series

Table 3: Handover comparison (seed 7, 300 samples)

Configuration	Handovers	Min serving RSRP	Samples < −100 dBm
Naive (0 dB, no TTT)	11	−102.7 dBm	4
Hysteresis only (3 dB)	5	−103.4 dBm	5
Hysteresis 3 dB + TTT 5	1 (sample 154)	−105.9 dBm	25

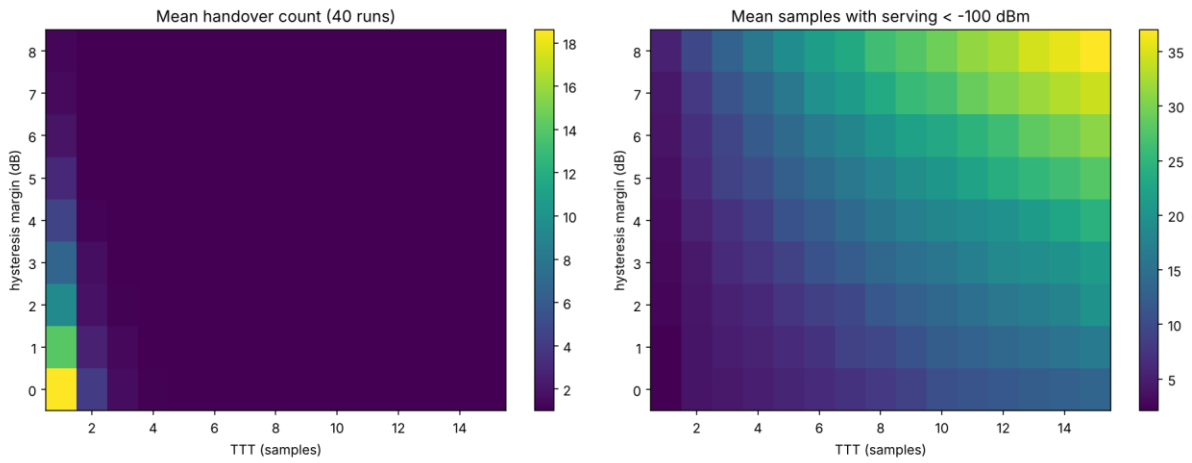


Figure 6: Margin and TTT sweep over 40 random runs (mean handover count and mean low-signal samples)

Table 4: Selected sweep results (mean of 40 runs)

Margin / TTT	Mean handovers	Mean samples < −100 dBm
0 dB / 1	18.60	2.1
3 dB / 1	6.75	2.8
3 dB / 5	1.00	9.0
6 dB / 10	1.00	23.1
8 dB / 15	1.00	37.0

2.4 Discussion

Q = √(3N) rises with the reuse factor: 1.73 for N = 1, 3.00 for N = 3, 3.46 for N = 4 and 4.58 for N = 7, and the generated layouts confirm that the nearest same-numbered hexagons sit exactly √(3N)·R apart. Small N packs co-channel cells close together, which gives high capacity but high co-channel interference; N = 7 separates them most, at the cost of reusing each frequency in fewer cells. The extension to N = 9 and N = 12 (Q = 5.20 and 6.00) continues the same trend with diminishing capacity.

In the handover simulation the naive trigger produced 11 handovers, nearly all between samples 105 and 145 where the two RSRP curves overlap. This is the ping-pong effect: noise keeps flipping which cell is momentarily stronger. A 3 dB hysteresis margin cut the count to 5, and adding a 5-sample time-to-trigger reduced it to a single handover at sample 154.

The price is the too-late-handover risk. With hysteresis plus TTT the serving RSRP stayed below −100 dBm for 25 samples (minimum −105.9 dBm), against 4 samples for the naive trigger, because the device waited longer on a fading serving cell. The sweep shows the trade-off worsening for aggressive settings: 8 dB with 15 samples gave about 37 low-signal samples on average, which is how over-tuning leads toward radio link failure. The 40-run averages (18.6, 6.75 and 1.00 handovers) show the same ordering as the single seeded run.

3. Mini Project 3.2: Cellular Capacity and Frequency-Reuse Planning

3.1 Objective

To compute system capacity for several reuse factors, show it together with Q, quantify cell splitting as a capacity lever, and illustrate how sectoring allows tighter reuse without a proportional interference increase.

3.2 Inputs and method

Table 5: Input parameters

Parameter	Value
Total spectrum	25 MHz
Channel bandwidth	200 kHz (125 channels in total)
Coverage area	1000 km²
Cell radius R	2 km (hexagon area 10.39 km², 96 cells fit)
Candidate N	1, 3, 4, 7, 9, 12
Splitting	R divided by 2 and by 4, at N = 7
Sectoring model	SIR ≈ Q^γ / i_0, γ = 4; 6 interferers (omni), 2 (120° sector)

- Channels per cell is ⌊125 / N⌋ and the number of cells is the area divided by the hexagon area $(3\sqrt{3}/2)R^2$. System capacity is channels per cell times number of cells.

3.3 Results

```
=== Inputs ===
Total spectrum        : 25.0 MHz
Channel bandwidth     : 200.0 kHz
Total channels        : 125
Coverage area         : 1000.0 km^2
Cell radius R         : 2.0 km  -> hexagon area 10.392 km^2
Cells that fit        : 96

=== Capacity versus reuse factor (omni cells, R = 2.0 km) ===
  N |      Q | channels/cell |  cells | system capacity (channels)
--------------------------------------------------------------------
  1 |   1.73 |           125 |     96 |                      12000
  3 |   3.00 |            41 |     96 |                       3936
  4 |   3.46 |            31 |     96 |                       2976
  7 |   4.58 |            17 |     96 |                       1632
  9 |   5.20 |            13 |     96 |                       1248
 12 |   6.00 |            10 |     96 |                        960

=== Cell splitting, N = 7 ===
 R (km) |  cells | channels/cell |  capacity | gain vs base
-----------------------------------------------------------
   2.00 |     96 |            17 |      1632 |        1.00x
   1.00 |    384 |            17 |      6528 |        4.00x
   0.50 |   1539 |            17 |     26163 |       16.03x
(Q is unchanged by splitting: Q depends only on N, not on R.)

=== Sectoring illustration (path-loss exponent gamma = 4.0) ===
  N |     Q |  SIR omni (6 interferers) |  SIR 3-sector (2 interferers) | gain
-------------------------------------------------------------------------------
  1 |  1.73 |                    1.8 dB |                        6.5 dB | +4.8 dB
  3 |  3.00 |                   11.3 dB |                       16.1 dB | +4.8 dB
  4 |  3.46 |                   13.8 dB |                       18.6 dB | +4.8 dB
  7 |  4.58 |                   18.7 dB |                       23.4 dB | +4.8 dB
  9 |  5.20 |                   20.8 dB |                       25.6 dB | +4.8 dB
 12 |  6.00 |                   23.3 dB |                       28.1 dB | +4.8 dB

Omni N=7 SIR target = 18.7 dB (tolerance 0.5 dB)
Smallest candidate N meeting that SIR with 3 sectors: N = 4 (SIR 18.6 dB)
Capacity omni N=7: 1632  |  3-sector at N=4: 2976  (1.82x, before trunking-efficiency loss)

Outputs written to /home/claude/mc_unit3/out_3_2
```

Figure 7: Terminal output of mp3_2.py (capacity, cell splitting and sectoring tables)

Table 6: Capacity versus reuse factor (R = 2 km)

N	Q	Channels per cell	Cells	System capacity
1	1.73	125	96	12,000
3	3.00	41	96	3,936
4	3.46	31	96	2,976
7	4.58	17	96	1,632
9	5.20	13	96	1,248
12	6.00	10	96	960

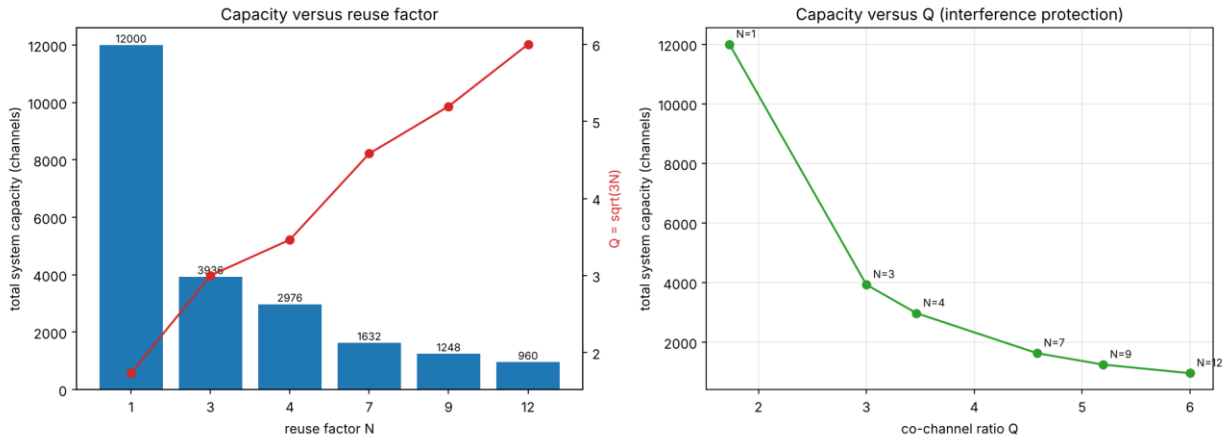


Figure 8: System capacity versus reuse factor N and versus Q

Table 7: Cell splitting at N = 7

R (km)	Cells	Channels per cell	Capacity	Gain
2.00	96	17	1,632	1.00×
1.00	384	17	6,528	4.00×
0.50	1,539	17	26,163	16.03×

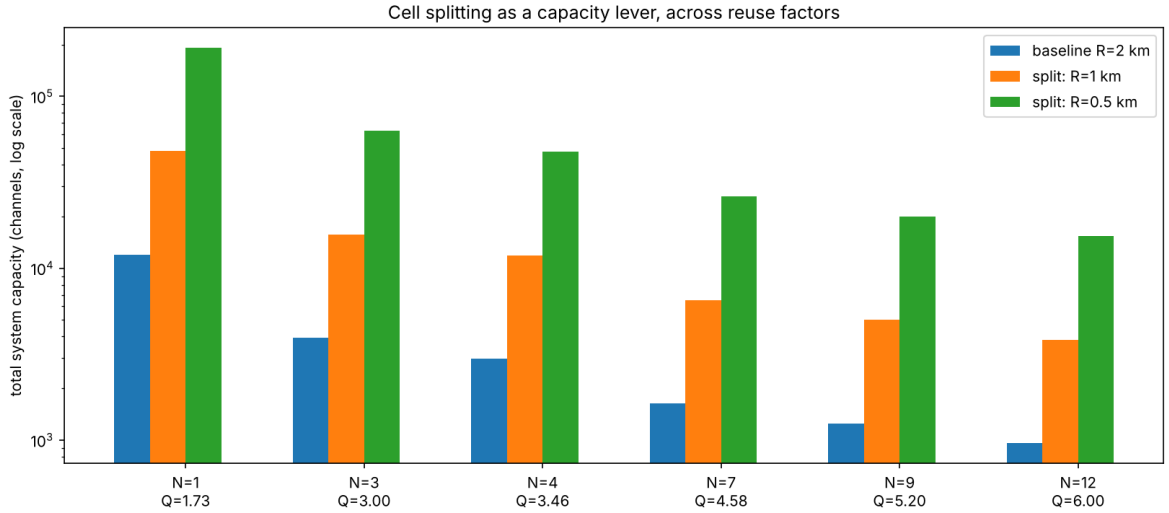


Figure 9: Cell splitting as a capacity lever across reuse factors (log scale)

Table 8: Sectoring illustration (γ = 4)

N	Q	SIR omni	SIR 3-sector	Gain
1	1.73	1.8 dB	6.5 dB	+4.8 dB
3	3.00	11.3 dB	16.1 dB	+4.8 dB
4	3.46	13.8 dB	18.6 dB	+4.8 dB

N	Q	SIR omni	SIR 3-sector	Gain
7	4.58	18.7 dB	23.4 dB	+4.8 dB
9	5.20	20.8 dB	25.6 dB	+4.8 dB
12	6.00	23.3 dB	28.1 dB	+4.8 dB

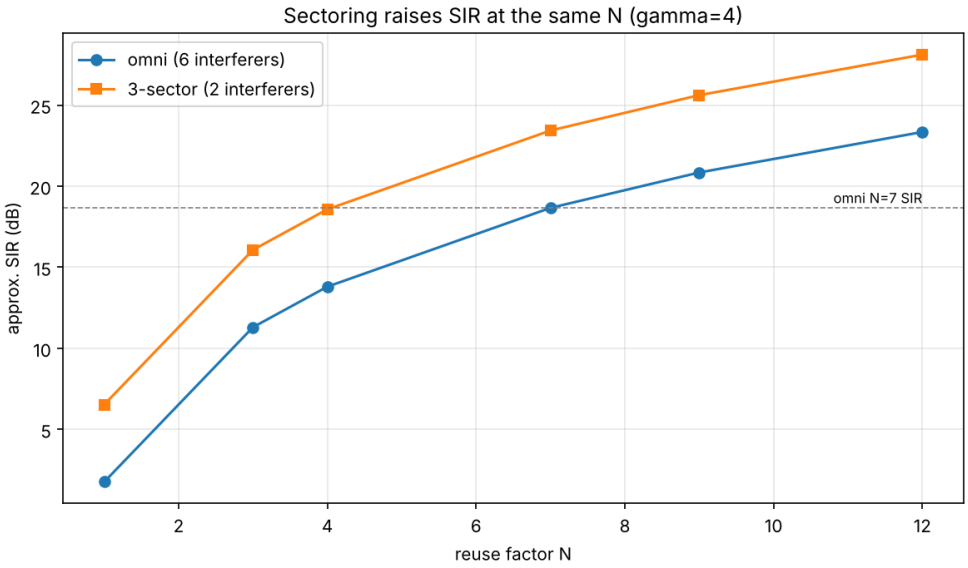


Figure 10: Approximate SIR versus N for omni and three-sector cells

3.4 Analysis

With a fixed 125 channels, capacity is inversely proportional to N: moving from N = 7 to N = 3 raises capacity from 1,632 to 3,936 channels, and N = 1 reaches 12,000. The price is co-channel protection, measured by Q = √(3N), which falls from 4.58 at N = 7 to 3.00 at N = 3 and 1.73 at N = 1. With a path-loss exponent of 4, omni N = 1 gives roughly 1.8 dB of SIR, which is unusable for narrowband channels, while N = 3 gives about 11 dB and N = 7 about 19 dB. Choosing N therefore means choosing the smallest value whose SIR the radio technology can tolerate.

For dense urban deployment, demand rather than coverage is the limit, so the operator wants the smallest workable N, typically 3 or 4 with sectored sites, combined with small cells. For rural deployment the opposite holds: traffic is light, cells are large and spectrum is not the bottleneck, so a larger N such as 7 buys robust SIR and range at no real cost.

Once N is already small, reducing it further is not available because dropping from N = 3 to N = 1 destroys SIR. Splitting and sectoring add capacity without that penalty. Cell splitting multiplies capacity by the square of the radius ratio (4× for halving R, about 16× for quartering it) and leaves Q, and therefore SIR, unchanged, because Q depends only on N. Sectoring works through the directional-beam argument of Section 3.1: a 120° antenna illuminates only a third of the surroundings and hears roughly two of the six first-tier co-channel cells, a 4.8 dB SIR gain. That gain lets N = 4 with three sectors (18.6 dB) match omni N = 7 (18.7 dB) while delivering about 1.8 times the capacity (2,976 against 1,632 channels).

Both levers have costs. Splitting needs new sites, backhaul and re-planning, and smaller cells mean more handovers, which is why the handover tuning in Project 3.1 matters more as cells shrink. Sectoring divides each site's channel pool across three sectors, which reduces trunking efficiency, and adds sector-to-sector handovers. The model counts channels, ignores traffic (Erlang) dimensioning, uses a first-tier-only SIR approximation and assumes perfect hexagons, so the absolute numbers are indicative and the trends are the point.

4. Mini Project 3.3: Connection and Handover as a Finite-State Machine

4.1 Objective

To model a device's connection lifecycle (Idle, Random Access, Connected, Handover in progress) as an explicit state machine, represent the four RACH steps, include a Radio Link Failure (RLF) path, and compare configurable handover policies on a scripted three-cell scenario.

4.2 Design

- **States:** IDLE; RACH with four sub-states (preamble, Random Access Response with timing advance, connection request, contention resolution); CONNECTED; HANDOVER.
- **Policies:** interchangeable functions policy(serving, rsrp, mem) that return a target cell or none. The naive policy hands over when any neighbour is stronger; the hysteresis plus TTT policy requires a margin held for a number of consecutive samples.
- **RLF:** declared when serving RSRP is below −110 dBm for 3 consecutive samples, in CONNECTED or HANDOVER. The device returns to IDLE, waits 5 samples, then re-attaches through RACH.
- **Scenario:** cells A, B and C at 0, 1000 and 2000 m on a straight road; device at 20 m/s from 50 m to 2000 m; 0.2 s samples (487 samples); path loss $-45 - 33 \cdot \log10(d/10)$ dBm; AR(1) shadowing ($\sigma = 3$ dB, $\rho = 0.9$) plus 1.5 dB measurement noise. The first attach is scripted to lose contention once so the RACH retry path appears in the log.

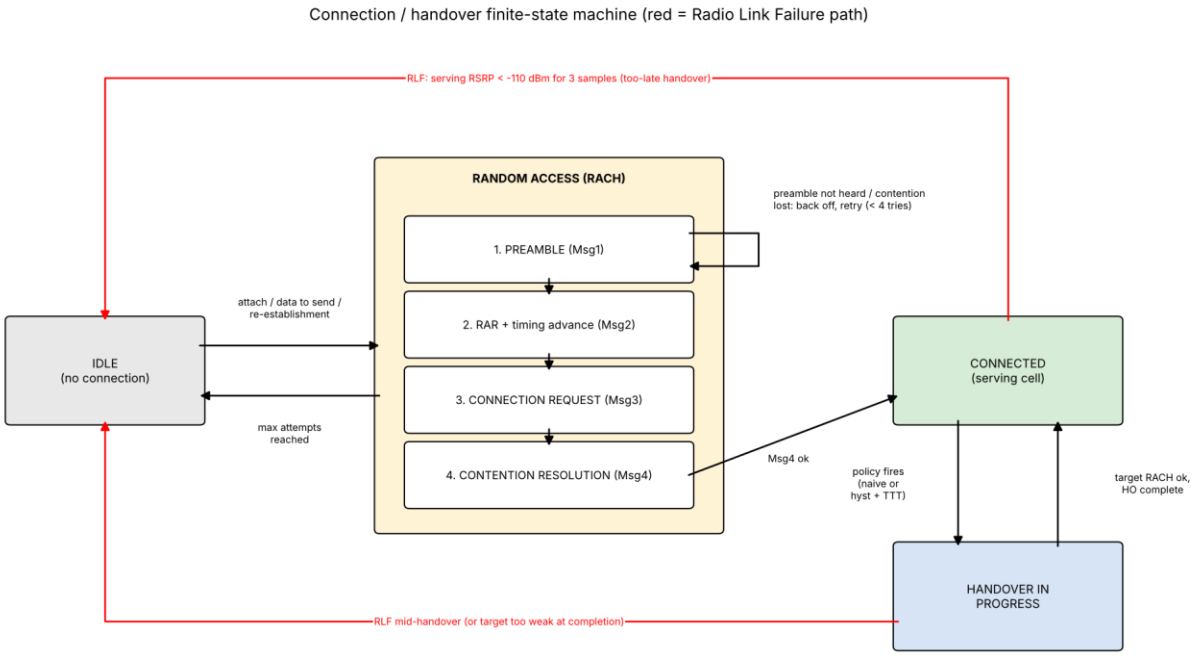


Figure 11: State-transition diagram (red = Radio Link Failure path)

Table 9: State-transition table

From	To	Trigger
IDLE	RACH_PREAMBLE	Data to send or re-establishment; strongest cell ≥ −105 dBm (5-sample wait after RLF)
RACH_PREAMBLE	RACH_RAR	Preamble detected (cell RSRP ≥ −110 dBm)
RACH_PREAMBLE	IDLE	4 attempts without detection
RACH_RAR	RACH_CONNREQ	RAR with timing advance received
RACH_CONNREQ	RACH_CONTENTION	Msg3 (RRC connection request) sent
RACH_CONTENTION	CONNECTED	Contention resolved (Msg4)
RACH_CONTENTION	RACH_PREAMBLE	Contention lost, back off and retry
CONNECTED	HANDOVER	Policy fires (naive, or margin held for TTT samples)
HANDOVER	CONNECTED	Execution done (3 samples) and target RSRP ≥ −110 dBm
CONNECTED	IDLE	RLF: serving RSRP < −110 dBm for 3 consecutive samples
HANDOVER	IDLE	RLF on source during execution, or target too weak

4.3 Results

```
Scenario: 487 samples x 0.2s = 97 s, device 20.0 m/s from x=50 to 2000 m
Cells: {'A': 0.0, 'B': 1000.0, 'C': 2000.0};  RLF below -110 dBm for 3 samples

Policy                              | Handovers | RLF | Ping-pong | Attaches
-------------------------------------------------------------------------------
naive (no margin, no TTT)           |        16 |   0 |        13 |        1
hysteresis 3 dB + TTT 5 samples     |         2 |   0 |         0 |        1
hysteresis 8 dB + TTT 60 samples    |         0 |   2 |         0 |        3

Log excerpt (hysteresis 3 dB + TTT 5):
  t=  0 IDLE            -> RACH_PREAMBLE    A   cell selection: A at -67.6 dBm; data to send -> send
      preamble (Msg1)
  t=  1 RACH_PREAMBLE   -> RACH_RAR         A   preamble detected -> Random Access Response (Msg2)
  t=  2 RACH_RAR        -> RACH_CONNREQ     A   RAR received, timing advance = 1 steps -> RRC connection
      request (Msg3)
  t=  3 RACH_CONNREQ    -> RACH_CONTENTION  A   Msg3 sent -> awaiting contention resolution (Msg4)
  t=  4 RACH_CONTENTION -> RACH_PREAMBLE    A   contention lost (identity mismatch) -> back off, new
      preamble
  t=  5 RACH_PREAMBLE   -> RACH_RAR         A   preamble detected -> Random Access Response (Msg2)
  t=  6 RACH_RAR        -> RACH_CONNREQ     A   RAR received, timing advance = 1 steps -> RRC connection
      request (Msg3)
  t=  7 RACH_CONNREQ    -> RACH_CONTENTION  A   Msg3 sent -> awaiting contention resolution (Msg4)
  t=  8 RACH_CONTENTION -> CONNECTED        A   contention resolved (Msg4) -> RRC connected
  t=146 CONNECTED       -> HANDOVER         A   policy fired: B -100.1 vs A -106.8 dBm -> handover command
      to B
  t=149 HANDOVER        -> CONNECTED        B   handover complete: A -> B (target RACH ok)
  t=382 CONNECTED       -> HANDOVER         B   policy fired: C -103.1 vs B -108.7 dBm -> handover command
      to C
  t=385 HANDOVER        -> CONNECTED        C   handover complete: B -> C (target RACH ok)

Outputs written to /home/claude/mc_unit3/out_3_3
```

Figure 12: Terminal output of mp3_3.py (policy summary and transition-log excerpt)

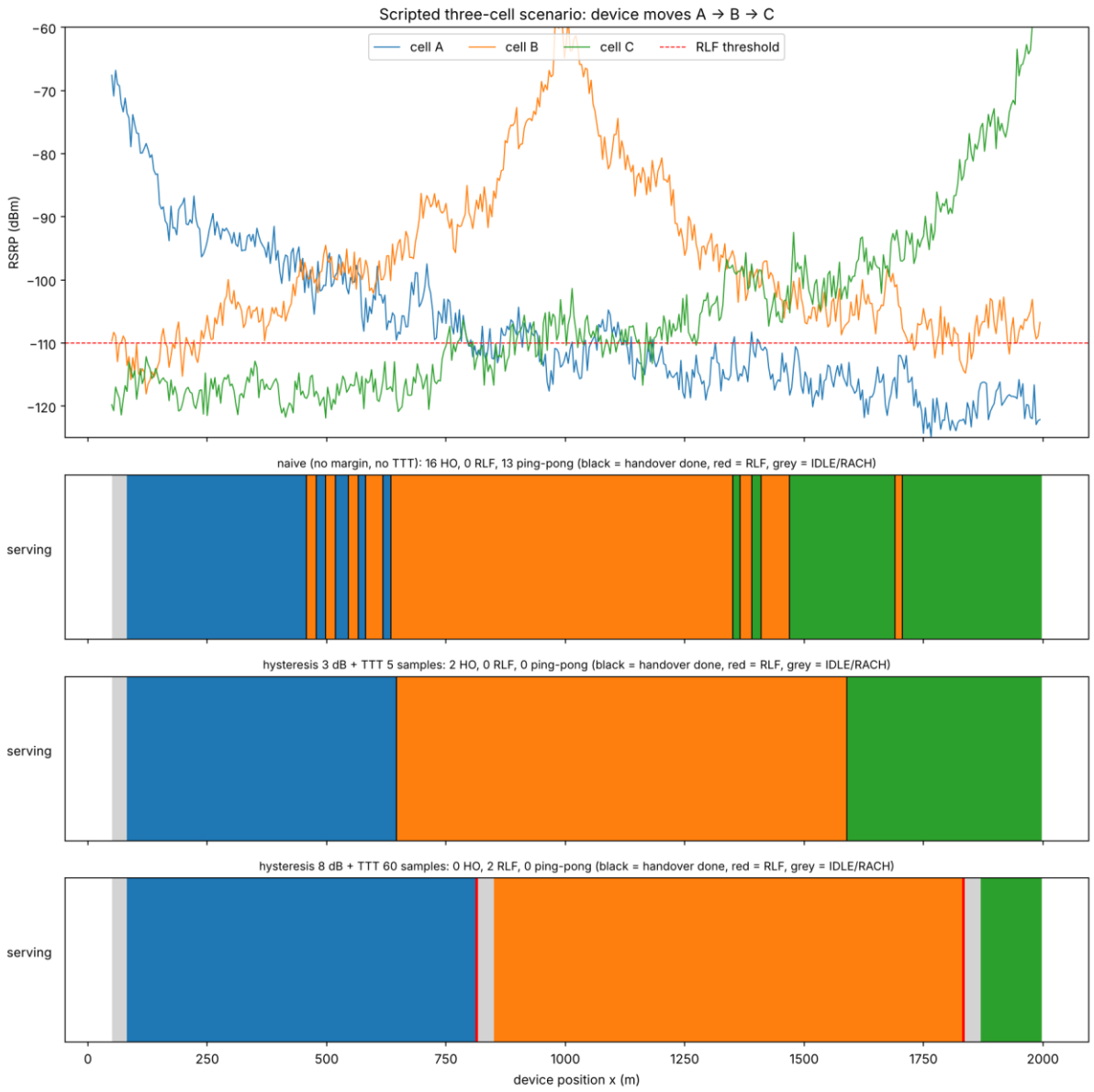


Figure 13: Generated signal series for cells A, B, C and the serving-cell timeline under each policy

Table 10: Per-policy results

Policy	Handovers	RLF	Ping-pong	RACH attaches
Naive	16	0	13	1
Hysteresis 3 dB + TTT 5	2	0	0	1
Hysteresis 8 dB + TTT 60	0	2	0	3

```
Policy: hysteresis 8 dB + TTT 60 samples
   7     1.4     78  RACH_CONNREQ    -> RACH_CONTENTION  A    Msg3 sent -> awaiting contention resolution (Msg4)
   8     1.6     82  RACH_CONTENTION -> CONNECTED        A    contention resolved (Msg4) -> RRC connected
 191    38.2    814  CONNECTED       -> IDLE             -    RADIO LINK FAILURE: A -111.9 dBm < -110 dBm for 3
      samples (too-late handover)
 196    39.2    834  IDLE            -> RACH_PREAMBLE    B    cell selection: B at -87.7 dBm; data to send -> send
      preamble (Msg1)
 455    91.0   1870  RACH_CONTENTION -> CONNECTED        C    contention resolved (Msg4) -> RRC connected

Handovers completed: 0   RLF: 2   Ping-pongs: 0
```

Figure 14: Transition-log excerpt for the over-tuned policy, showing both Radio Link Failures and re-attachment through RACH

4.4 Comparison of policies

The naive policy handed over whenever a neighbour was momentarily stronger. In the A/B overlap (x ≈ 450 to 650 m) and the B/C overlap (x ≈ 1340 to 1700 m) it completed 16 handovers, 13 of which were ping-pongs, meaning a return to a cell left within 8 seconds. It never lost the link because it always moved early, but it generated a large signalling load for no benefit.

Hysteresis of 3 dB with a 5-sample time-to-trigger removed all ping-pong. The margin ensures that a neighbour must be clearly better, and the TTT ensures that this holds for 1 second rather than for a single noisy sample. It produced exactly the two intended handovers, A to B at x = 634 m and B to C at x = 1578 m. These are 134 m and 78 m beyond the geometric midpoints, and the serving RSRP at trigger was −106.8 and −108.7 dBm, close to the −110 dBm RLF threshold but still above it. That shows the cost of this filtering: handovers become later.

The over-tuned setting (8 dB, 60 samples = 12 s) shows that cost taken too far. The required advantage never persisted for 60 consecutive samples before the serving cell fell below −110 dBm, so no handover happened and the device suffered two radio link failures, at x = 814 m and x = 1834 m. Each caused about 1.8 seconds of outage while the device waited in IDLE and re-attached through RACH. This is the too-late handover failure.

A larger margin and a longer TTT reduce ping-pong but delay handover, and both must be chosen against the fading and the device speed. Here 3 dB with 5 samples sat in the working range. This is a single seeded scenario, so the precise counts would change with another seed even though the ordering is stable.

5. Conclusion

- Q = √(3N) grows with N, so smaller reuse factors give more capacity and less co-channel protection. Simulated layouts matched the formula exactly.
- Capacity falls from 12,000 channels at N = 1 to 960 at N = 12 for the chosen inputs. Cell splitting multiplies capacity by the square of the radius ratio without changing Q, and three-sector sites let N = 4 match the interference quality of omni N = 7 with about 1.8 times the capacity.
- Naive handover causes ping-pong in overlap regions. Hysteresis and TTT remove it, but too much of either delays handover and leads to radio link failure, so the parameters must be tuned, which is the task of mobility robustness optimisation.
- The connection state machine with RACH, handover and RLF paths reproduced these behaviours in a three-cell mobility scenario.